

Dual Degree Project Entitled

Simulations on packed beds

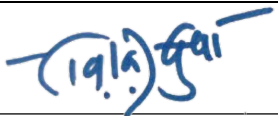
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	08 June 2022

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Abstract:

Before carrying out fluid flow and heat-mass transfer related simulations to characterize packed beds, one needs to be able to generate the bed either by carrying out a physical experiment and replicating the packing in Ansys or by using some other computational tools to synthetically generate the bed. Here, we use Blender as a tool to model our synthetic bed. Blender is an open source software running on Python which can be used for rigid body simulations. The beds thus generated are then further post processed to generate porosity data using image processing tools.

Next, simulations of fluid flow were carried out to understand the effect of different foam bed geometries on the pressure and velocity distributions through the bed. For this purpose, the Kelvin foam bed geometry was created and the flow profiles compared using three different RANS turbulence models (k - ϵ , k - ω , and SST k - ω).

Further, a foam bed geometry from a bed of spherical particles was created and the simulated results compared for 1%, 5% and 10% volume expansion in the particles.

Next, heat transfer simulations were carried out for fluid flow through a packed bed of spherical particles with heating being provided through a sandwich of hot fluid. The source term is set at -50 kW/m^3 for the sandwiched cold fluid.

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Introduction:

Packed bed reactors find wide usage in the chemical and process industry. They facilitate gas-solid catalysed reactions, the likes of which include methanol synthesis, methane-steam reforming and ammonia synthesis. The high specific surface area provided by packed beds is what facilitates enhanced heat and mass transfer thereby increasing reaction rates inside the catalyst.

Instead of physically creating a bed and replicating the packing in a simulation software like Ansys, we use a much easier and appreciably reliable approach to obtain a bed packing i.e. we use an open source software for rigid body simulation viz. Blender. After bed generation is completed, the location and orientation data for the bed particles are extracted which are then used to generate images for planar impressions of the bed particles at different z-values (along the axis of the bed). Finally, image analysis is performed to obtain a z-averaged porosity profile data. This study aims at creating beds with particles of different shapes and comparing their porosity profiles. Moving forward, these beds will be compared for the effect that change in particle shape/dimensions has on the heat-mass transfer phenomenon using a simulation tool like Ansys.

The catalyst shape and the tube to particle diameter are two important factors that play a role in influencing the bed performance viz. pressure drop, catalyst effectiveness, heat transfer, and mechanical strength. In particular, catalytic foams are being studied because of their characteristically high specific surface area. It is worthwhile to note that foam beds use wash-coated catalysts i.e. the catalyst is coated only on the surface of the foam. Despite the reduced diffusional resistance, this reduces the catalyst volume inside the reactor. Thus, it would be interesting to look at the possibility of using a bulk foam catalyst bed.

The objective of this work is then to also look at two different foam bed geometries viz. (a) the Kelvin cell foam bed and (b) the spherical particle foam bed. Apart from looking at the flow profiles for these geometries, comparison is also made between the z-averaged radial porosity profiles for the beds generated using a spherical particle packing for 1%, 5% and 10% increase in the particle volume.

Finally, to quantify the effect of contact surface type on heat transfer characteristics, relatively cold air was heated with a sandwich of hot air with the contact surface b/w the hot-cold regions varied between a) a planar boundary & b) zig-zag corrugated boundary.

Modelling/Experiment:

A. Bed generation

Blender is an open source software which works on Python. Python scripts were written for automating bed generation for the following particles:

1. Cylcut
2. Cylinder
3. Daisy
4. Cylinder with internal and external holes
5. 7-hole
6. Quad-lobe
7. Sphere
8. Tri-lobe
9. 7-hole with bulge

Two different forms of the script were written for dropping the particles viz. dropping them all at once and secondly, keyframing them so that they drop one at a time.

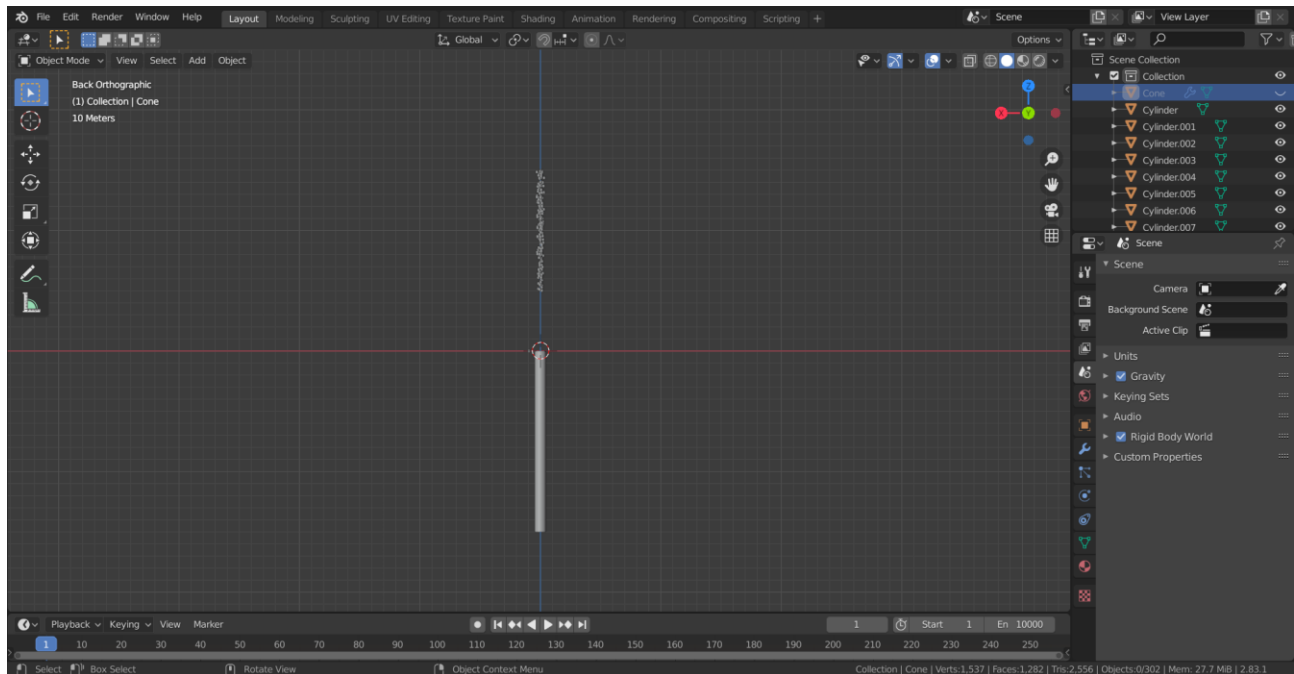


Figure 1. Version 1: Particles dropped all at once

(Note: Particles above, tube below)

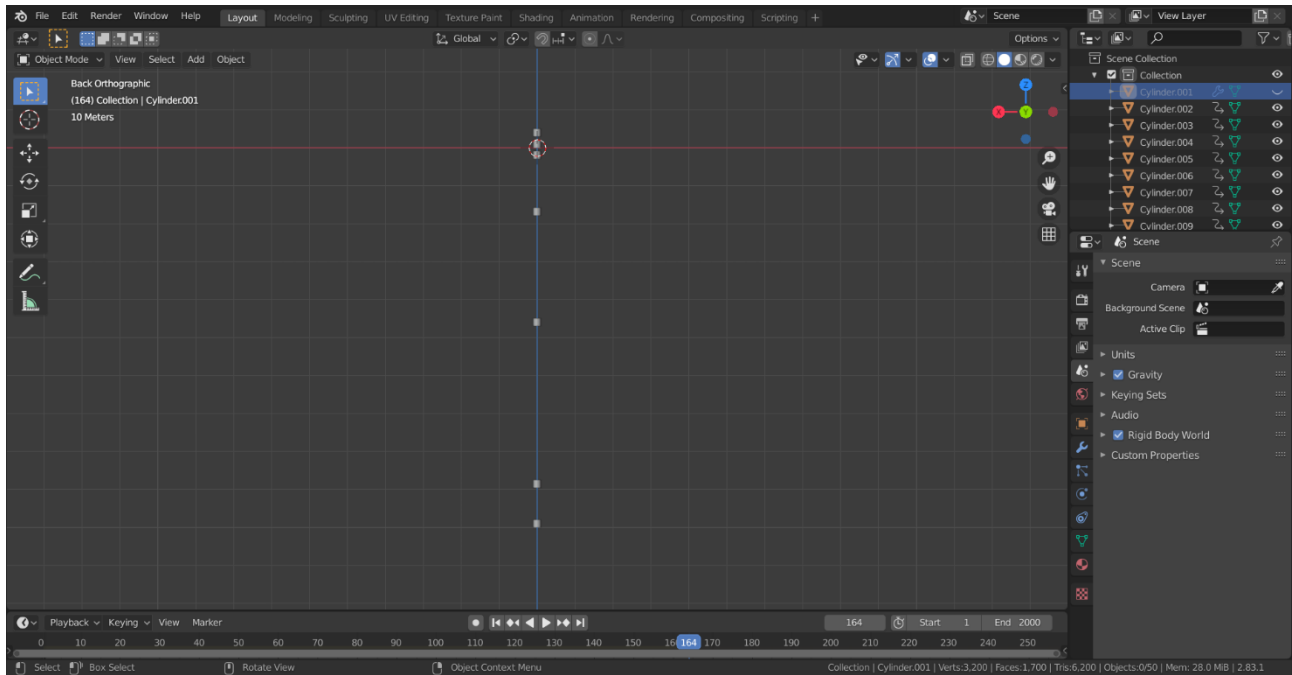


Figure 2. Version 2: Particles dropped one at a time

(Note: Particles being shown dropped one at a time, tube omitted for clarity)

Finally, we decided to stick to filling the bed by dropping particles one at a time in all further work to remain consistent.

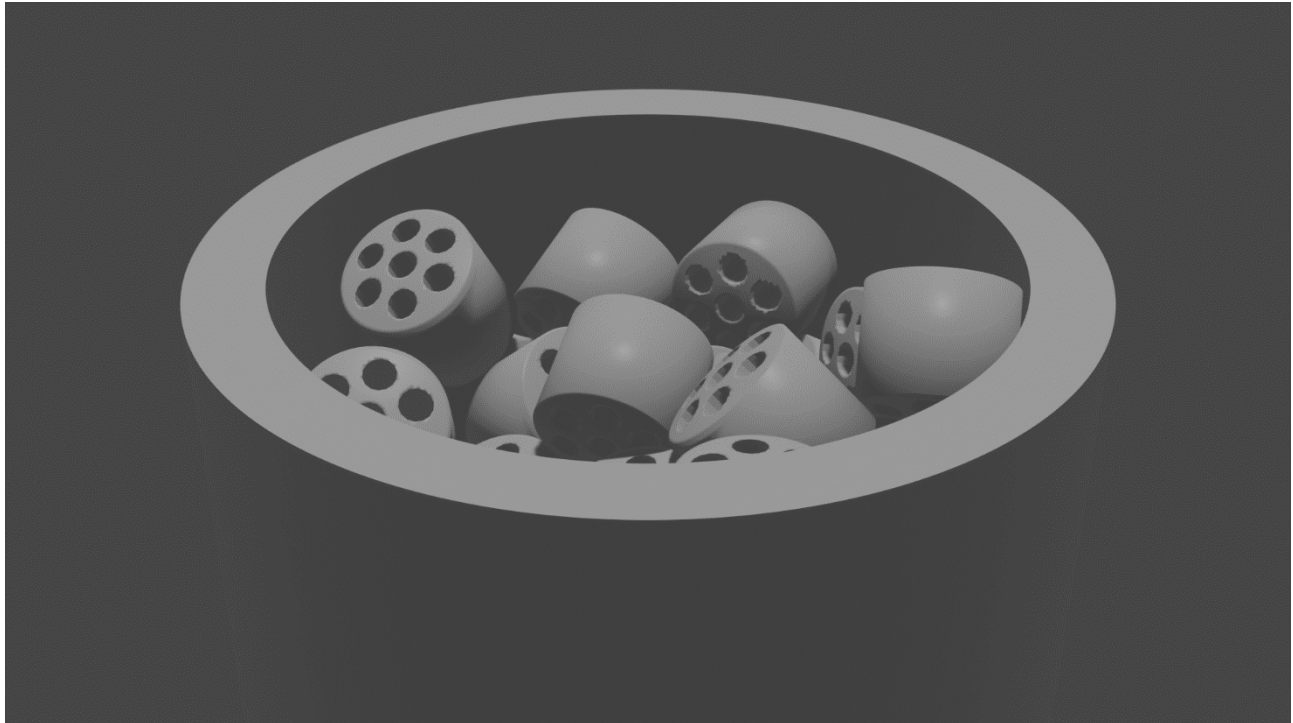


Figure 3. Generated bed after rigid body simulation is complete

B. Image generation

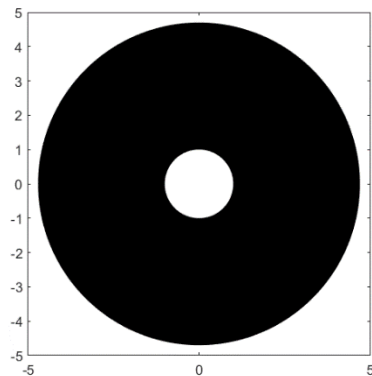
Location and orientation data from the generated bed is extracted, then fed into code written in MATLAB for image generation. Separate codes have been written for each shape of the particle for the purpose. The codes work on the following basic geometrical concepts:

1. Projection of a cylinder on a plane is elliptical with the minor axis equal to the diameter of the cylinder and the major axis equal to $d \times \sec \theta$ where θ is the angle made by the axis of the cylinder with the normal to the plane. The direction of the major axis is along the projection of the cylinder axis on the plane and the minor axis is along the line perpendicular to it. The center of the ellipse is found by the intersection of the cylinder axis with the equation of the plane.
2. Projection of a plane on another plane is a line directed along the cross product of the normals to the two given planes.
3. Projection of a sphere on a plane is simply a circle, the diameter of which is governed by the distance of the sphere's center from the plane.

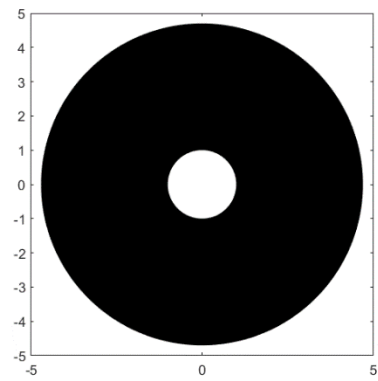
Now, we note that the outer surfaces of all of the particles considered so far can be considered to be built up from a combination of a cylinder/part-cylinder, a plane and a sphere/part-sphere.

Using the above three geometrical concepts individually on all the surfaces that build up the boundary of a particle, we arrive at the projection of the particle on a plane. Scripted in MATLAB, the above concepts give out the following results for the projections of particles on planes:

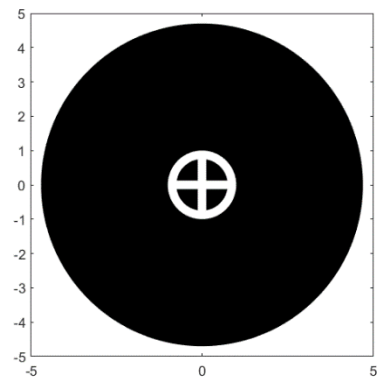
1. Cylinder:



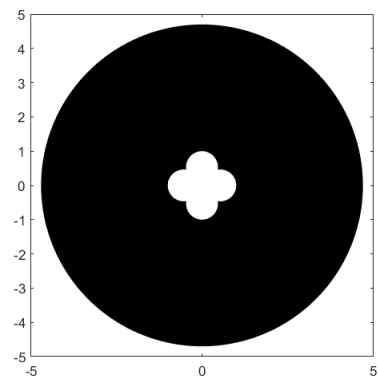
2. Sphere:



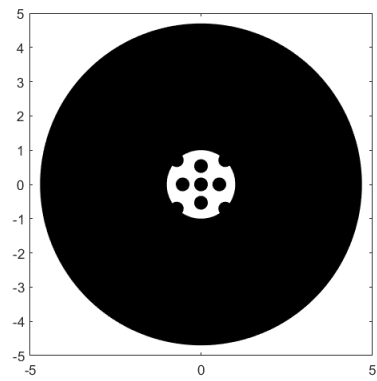
3. Cylcut:



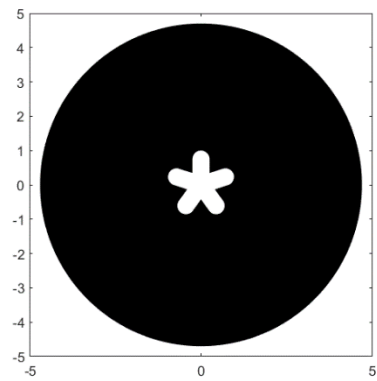
4. Quad-lobe:



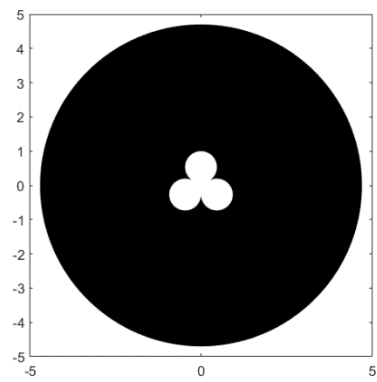
5. Cylinder with internal and external holes:



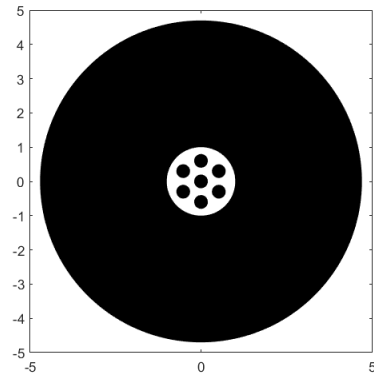
6. Daisy:



7. Tri-lobe:



8. 7-hole:



Next, we found that it was possible and much easier to generate images of particle impressions on planes using Blender itself by use of the Boolean modifier. We create planes parallel to the bed bottom and apply the Boolean Difference operation on the planes using all of the particles in the bed as the target object. This process is automated by writing a script in Python which also takes care of exporting and saving the images of the planes thus generated.

Hence, we eliminated using MATLAB for the image generation process and saved ourselves from writing a new particle specific script for each new particle that we encounter - the Blender script for image generation works universally for all particle shapes.

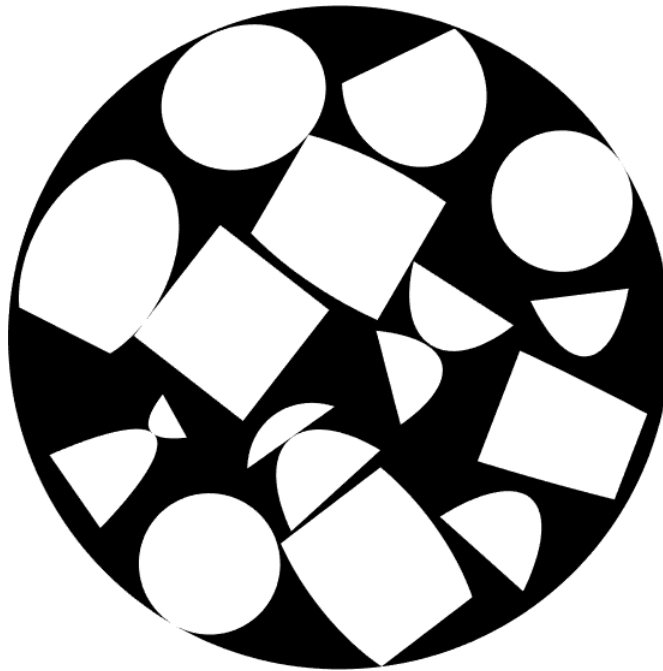


Figure 4. Version 1: Image generation using particle-specific script written in MATLAB



Figure 5. Version 2: Image generation using universal script written in Blender

C. Image analysis/porosity profile generation

Finally, the images thus generated are processed using a script written in MATLAB. First the images are binarized into a black and white form. Then, at a particular value of distance r from the center of the circular plane, we count the number of white and black pixels using the `impixel` tool. The ratio of the number of black pixels to the total number of pixels gives us the value of the porosity at that radial distance for a particular plane. Averaging this value over all planes gives us the z -averaged radial porosity profile for a particular bed.

D. Foam bed

D1. Kelvin cell foam bed

The Kelvin cell foam bed geometry used for the packed bed creation was inspired from this historical question posed by Lord Kelvin in 1887: *What three-dimensional arrangement of space-filling cells of equal volume has minimal surface area?* The problem cited is relevant to the current work as the dry foam has an energy proportional to the surface area of the films.

The solution posed by Lord Kelvin was a bi-truncated cubic honeycomb lattice or simply put a lattice of tetrakaidecahedra arranged in a body-centred cubic fashion. For more than a century, nobody could improve on Kelvin's solution to obtain a better surface to volume ratio until in 1993, when researchers Robert Phelan and Denis Weaire constructed a structure using two kinds of equal volume cells to arrive at what is now known as the Weaire–Phelan structure.

For the current study, Kelvin's solution is being used to model the foam geometry. The edges of this geometry serve as the skeleton of the foam. The strut geometry chosen in this study is cylindrical with a diameter of 3.38 mm and a particle diameter of 16.8 mm.

The geometry creation part of the process is done in Ansys SpaceClaim. A tetrakaidecahedron unit cell (see Figure 6) is first created which is then replicated in all three directions to give, after some boolean cut/intersect operations with a hollow cylinder created separately, a 3D Kelvin cell lattice (see Figure 7).

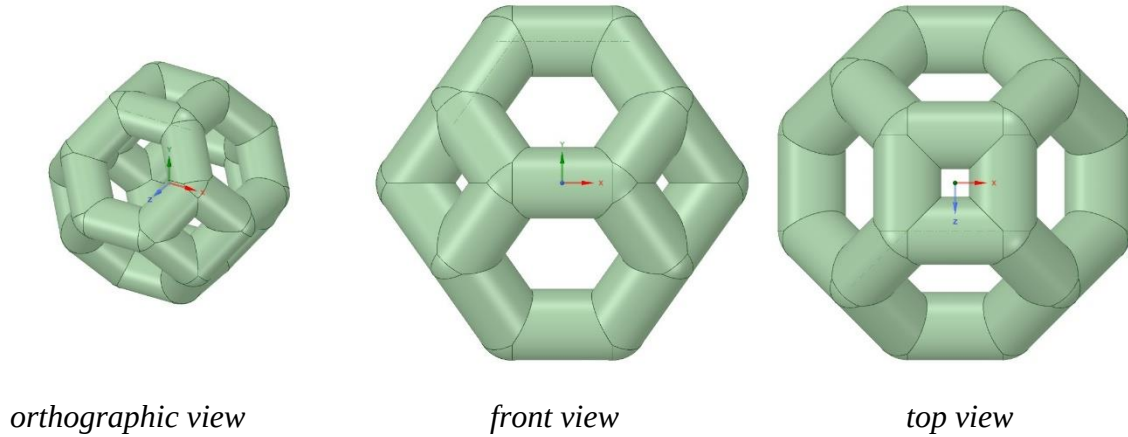


Figure 6. Kelvin unit cell

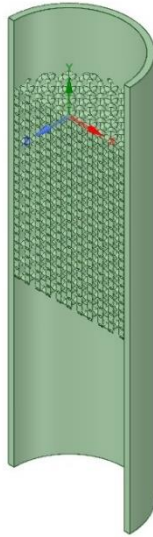


Figure 7. Kelvin cell lattice

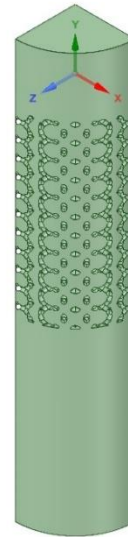


Figure 8. Fluid domain

The thus obtained 3-D lattice is used to create the geometry of the fluid domain (see Figure 8), only a quarter of which is used in the simulation. This reduces the computational effort by working under the assumption that the flow in the left-out regions is symmetric to that in the working region.

Next, meshing is performed on the thus obtained geometry using the Workbench utility (see Figure 9). Some mesh statistics for the thus generated mesh are given below:

Nodes	1784258	
Elements	9058921	
Element Quality	Average	0.837
	Minimum	0.144
	Maximum	0.955
Aspect Ratio	Average	2.24
	Minimum	2.2
	Maximum	21
Skewness	Average	0.229
	Minimum	0.0498
	Maximum	0.947

Table 1. Mesh Metrics for Kelvin cell foam bed mesh

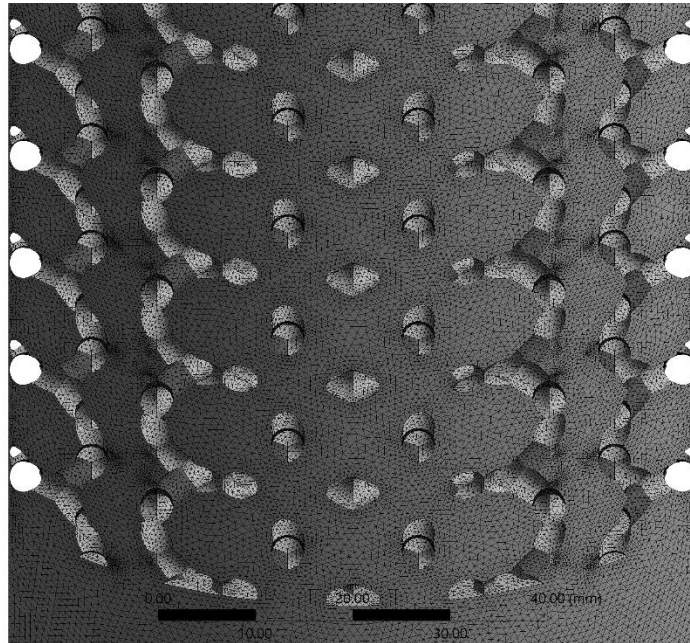


Figure 9. Meshed foam bed geometry

The simulation is performed using three different RANS turbulence models viz. (a) the $k-\varepsilon$ model, (b) the $k-\omega$ model, and (c) the SST $k-\omega$ model with air as fluid, Reynolds number of ≈ 50000 with symmetry boundary conditions at the cut walls and wall boundary conditions at the curved walls. A pseudo transient formulation has been used to be able to obtain a steady state profile of the turbulent simulations with relatively reduced CPU time.

The model equations for the RANS models is obtained by averaging the Navier-Stokes and continuity equations over appropriately small time intervals:

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot [\mu(\nabla \mathbf{U} + (\nabla \mathbf{U})^T)] + \rho g - \nabla \left(\frac{2}{3} \mu (\nabla \cdot \mathbf{U}) \right) - \nabla \cdot (\overline{\rho \mathbf{U}' \mathbf{U}'})$$

In order to calculate the turbulent stresses, some models are needed. One way is by using the Boussinesq hypothesis:

$$-\overline{\rho \mathbf{U}' \mathbf{U}'} = \mu_t (\nabla \mathbf{U} + (\nabla \mathbf{U})^T) - \frac{2}{3} \rho k \mathbf{I} - \frac{2}{3} (\nabla \cdot \mathbf{U}) \mathbf{I}$$

In order to close these equations, μ_t needs to be calculated. This is essentially where the k- ϵ , k- ω , and the SST k- ω models differ.

$$\begin{aligned} \frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \mathbf{U} k) &= \nabla \cdot \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right) + P_k - \rho \epsilon \\ \frac{\partial(\rho \epsilon)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \epsilon) &= \nabla \cdot \left(\left(\mu + \frac{\mu_t}{\sigma_\epsilon} \right) \nabla \epsilon \right) + C_1 \frac{\epsilon}{k} (P_k + C_3 P_b) - C_2 \rho \frac{\epsilon^2}{k} + S_c \end{aligned}$$

k- ϵ model equations

$$\begin{aligned} \frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho \mathbf{U} k) &= \nabla \cdot \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla k \right) + P_k - \rho \epsilon \\ \frac{\partial(\rho \omega)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \omega) &= \nabla \cdot \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla \omega \right) + \frac{\gamma}{\nu_t} P_k - \beta \rho \omega^2 \end{aligned}$$

k- ω model equations

Historically, the SST k- ω model was developed to tackle the drawbacks associated with the k- ϵ and k- ω model equations in certain situations. The k- ϵ model fails at accurately predicting adverse pressure gradients. The reason behind, being widely cited, that the damping functions inherent in the model equations are not very accurate near wall regions. Conversely, the k- ω model doesn't suffer from this drawback as the model equations inherently do not involve damping functions. However, the model has a drawback of it's own of being dependent on freestream turbulent conditions. In this respect, the k- ϵ model is better as it does not depend on the freestream turbulent conditions.

The SST k- ω appropriately takes advantage of the two models by switching to the k- ϵ model in the bulk region and to the k- ω model in the near wall region using a blending function, F_1 .

$$\begin{aligned} \frac{\partial(\rho \omega)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \omega) &= \nabla \cdot \left(\left(\mu + \frac{\mu_t}{\sigma_k} \right) \nabla \omega \right) + \frac{\gamma}{\nu_t} P_k - \beta \rho \omega^2 + 2(1 - F_1) \frac{\rho \sigma_{\omega 2}}{\omega} \nabla k : \nabla \omega \\ F_1 &= 1 \text{ (k - } \omega \text{ model)} \\ F_1 &= 0 \text{ (k - } \epsilon \text{ model)} \end{aligned}$$

SST k- ω model equations

Additionally, a viscosity limiter equation is introduced into the model for better agreement with experimental results.

$$\mu_t = \frac{a_1 \rho k}{\max(a_1 \omega, SF_2)}$$

The **pseudo transient formulation** is used as an under-relaxation scheme to speed up convergence. It uses a virtual time stepping method which is calculated as a minimum of several different time scales (convective, gravitational, rotational, diffusion, compressible, etc.) using characteristics dimensions to calculate each of these.

$$\rho_P \Delta V \frac{\phi_P - \phi_P^{old}}{\Delta t} + a_P \phi_P = \sum_{nb} a_{nb} \phi_{nb} + b$$

The pressure-velocity coupling scheme is kept as Coupled, spatial discretization for pressure used is Second Order, while the momentum and turbulent specifications discretization schemes as QUICK.

D2. Spherical particle foam bed

First, a spherical particle fixed bed is synthetically generated using a Python script written in Blender. After bed generation is completed, the location and orientation data for the bed particles are extracted which is then used to recreate the bed in Ansys using a Python script on SpaceClaim. The particle volumes are increased by 1%, 5% and 10% while recreating the structure in Ansys to create a foam geometry.

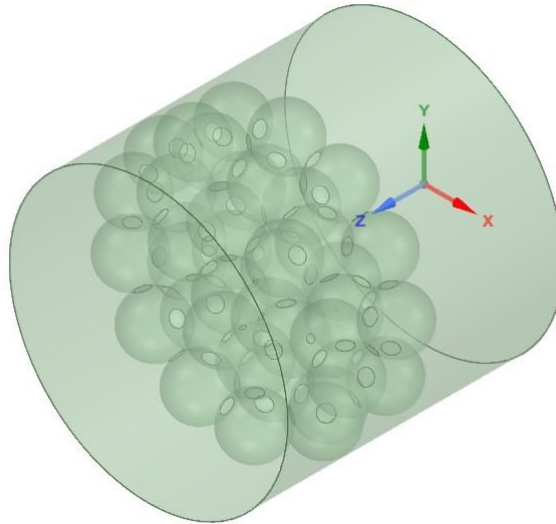


Figure 10. Spherical particle foam bed geometry

Mesh metrics for the geometry with particles volume expanded by 10% are given:

Nodes	56785	
Elements	247767	
Element Quality	Average	0.805
	Minimum	0.105
	Maximum	0.953
Aspect Ratio	Average	3.46
	Minimum	3.44
	Maximum	44.5
Skewness	Average	0.27
	Minimum	0.05
	Maximum	0.95

Table 2. Mesh Metrics for spherical particle foam bed mesh

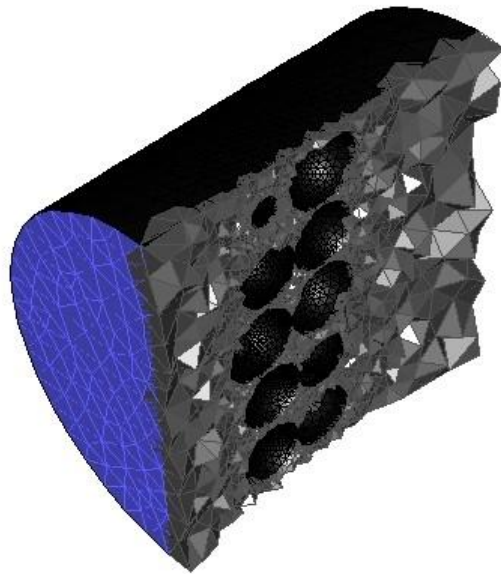


Figure 11. Cut section view of 10% bed mesh

Additionally, scripts are written in MATLAB to generate images of planar impressions of the bed particles at different z-values (along the axis of the bed). Image analysis is then performed using MATLAB to obtain a z-averaged porosity profile data.

The simulation is performed under laminar flow conditions with a tube based Reynolds number of 10. Pressure outlet boundary condition is used at the outlet with zero gauge pressure. The pressure-velocity coupling scheme is kept as SIMPLE, spatial discretization for pressure used is Second Order, while for momentum it is QUICK.

Next, before being able to perform a heat transfer simulation on the bed, some changes on the geometry are required to allow for the input of heat transfer flux into the bed and avoid any temperature spikes in sharp geometrical spaces.

Thus, the following two changes are made to the geometry:

1.

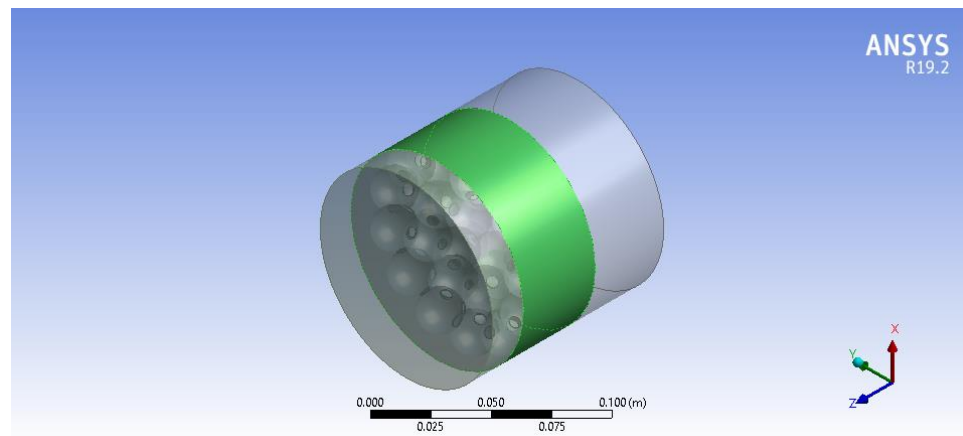


Figure 12. Heat flux into bed through green surface

The curved wall is divided into three regions wherein the highlighted region would serve as a source of heat flux into the bed.

2.

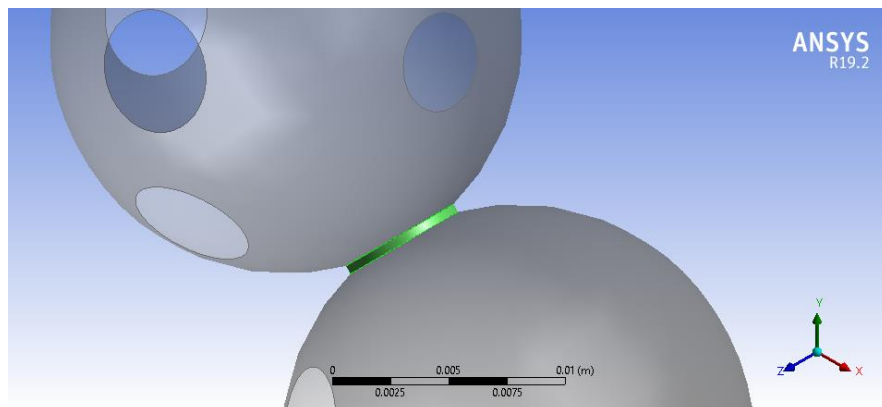


Figure 13. Chamfered contact b/w two spheres

Secondly, to prevent abrupt temperature spikes around sharp geometrical spaces, the sphere-to-sphere contact surfaces are chamfered to a specified thickness.

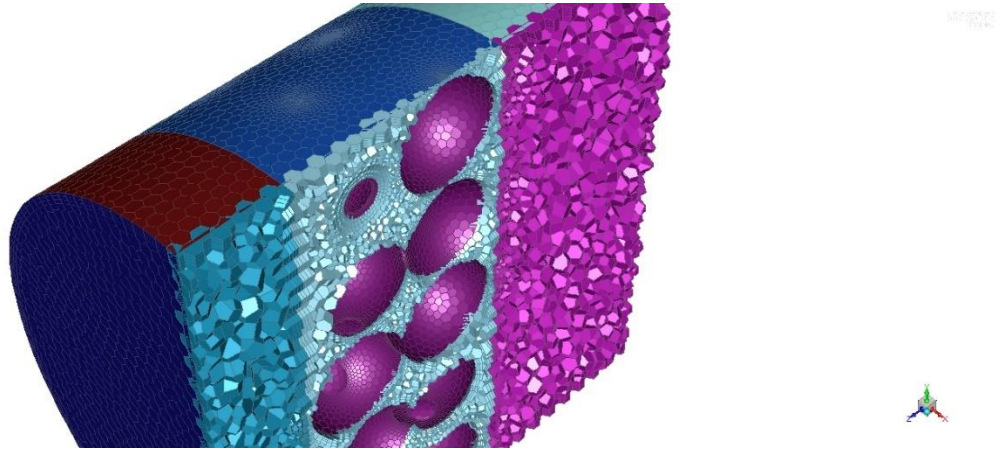


Figure 14. Cut plane through polyhedral meshed geometry

The resulting mesh created of the modified geometry is shown in Figure 14. The Fill With scheme is set as polyhedral with no Boundary Layers and the resultant mesh quality is reported in Table 3.

Nodes	11128068	
Cells	3855009	
Element Quality	Minimum	0
	Maximum	0.7999604
Aspect Ratio	Minimum	1.356318
	Maximum	22.22024
Skewness	Minimum	0
	Maximum	0.7999604

Table 3. Mesh Metrics for spherical particle foam bed mesh w/ modified geometry

The heat transfer simulation is performed under laminar flow conditions with air as the working fluid. The inlet boundary conditions are set such that the Reynolds number is ≈ 10 with a temperature of 300 K. The inlet flux through the conducting walls is set at 20 W/m^2 . The pressure-velocity coupling scheme is kept as SIMPLE, spatial discretization for pressure used is Second Order, while for momentum and energy it is QUICK.

E. Packed bed

To study the effect of contact surface corrugations on the heat transfer characteristics of packed beds, two different spherical particle packed bed geometries are created only differing in the contact surface type separating the fluid-fluid regions. Before generating the bed geometry in Ansys SpaceClaim, the packing of spherical particles is created in Blender using the Python script written earlier (see Figure 15). Later, the synthetically generated spherical particle arrangement is recreated in SpaceClaim along with the flow regions. The thickness of the fluid-fluid contact surfaces will later be set as 1 mm during case setup. The two geometries thus created are shown below in Figures 16 & 22, while the boundaries are identified in Figures 17-21:

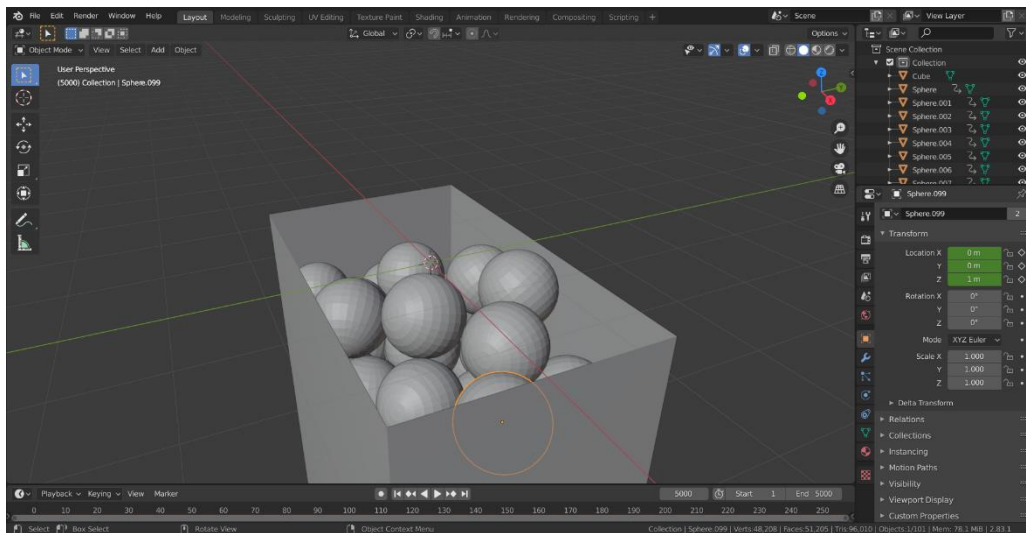


Figure 15. Blender generated synthetic packed bed

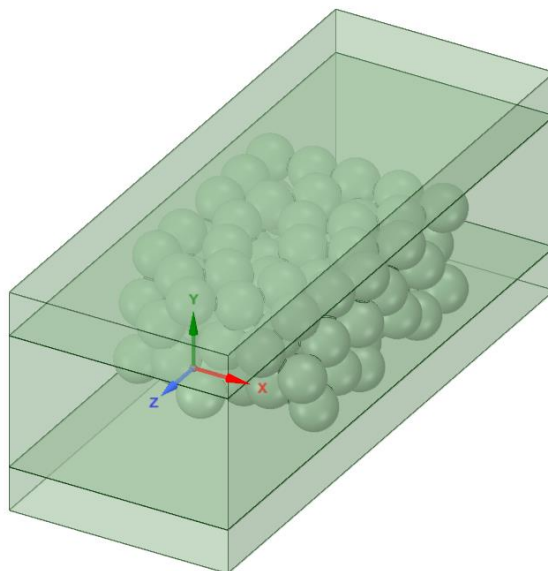


Figure 16. Spherical particle packed bed recreated in Ansys SpaceClaim

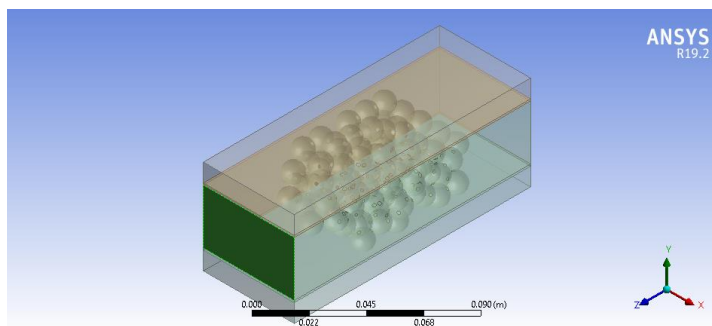


Figure 17. Gas inlet

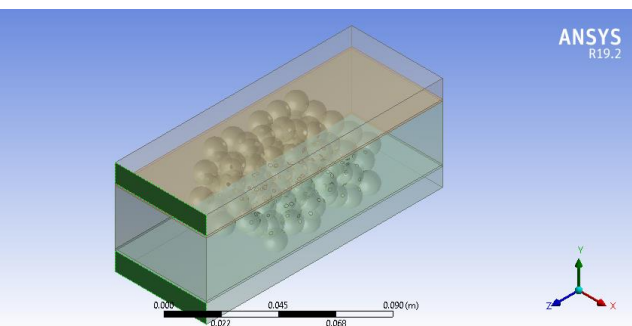


Figure 18. Superheated steam inlet

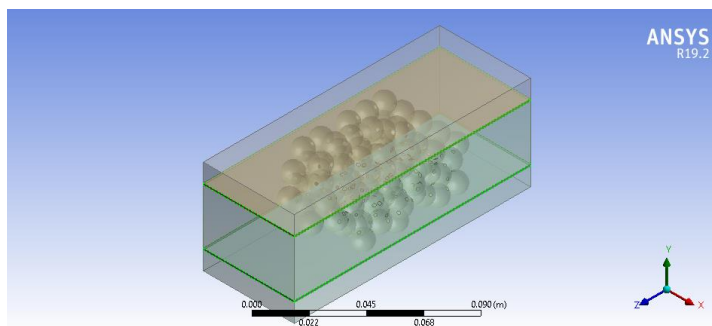


Figure 19. Conducting walls

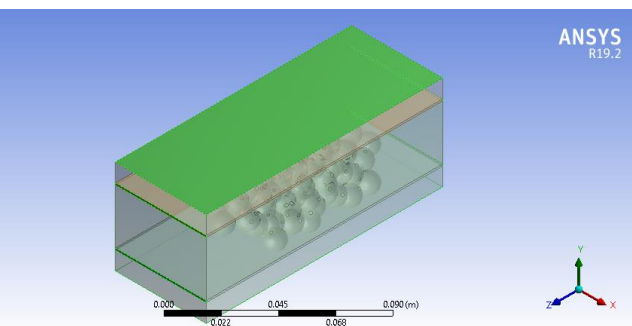


Figure 20. Insulated walls

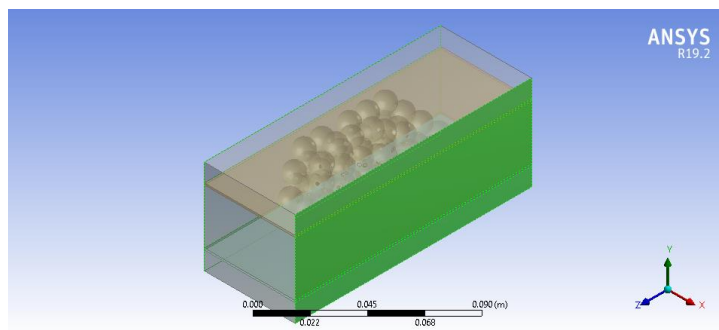


Figure 21. Symmetry walls

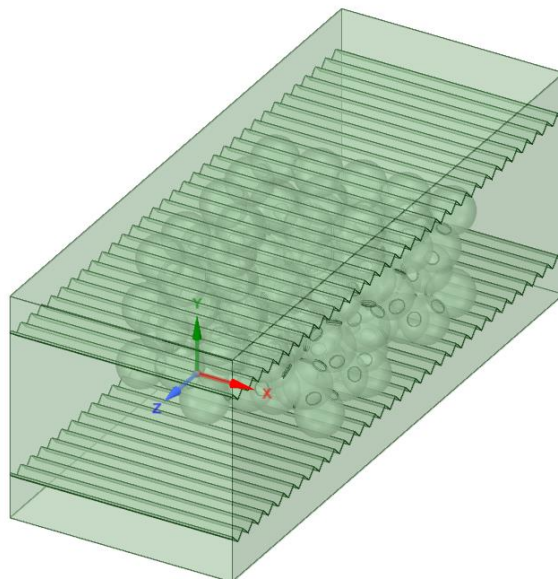


Figure 22. Spherical particle packed bed with corrugated fluid-fluid contact surface

Next, the geometries are meshed using the Fluent meshing utility to give the following meshes with associated metrics as shown in Tables 4 & 5 respectively. The Fill With scheme is set as poly-hexcore which automatically switches between structured tetrahedral meshing and polyhedral meshing as it seems fit.

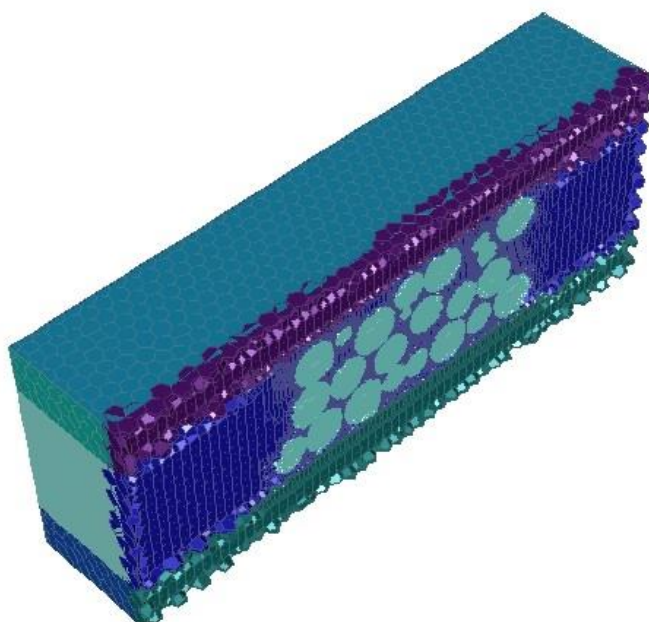


Figure 23. Meshing on geometry with planar fluid-fluid contact separation

Nodes	6860368	
Cells	1884842	
Element Quality	Minimum	1.337541
	Maximum	33.81832
Aspect Ratio	Minimum	1.337541
	Maximum	33.81832
Skewness	Minimum	0
	Maximum	0.7997312

Table 4. Mesh Metrics for packed bed w/ planar fluid-fluid contact surface

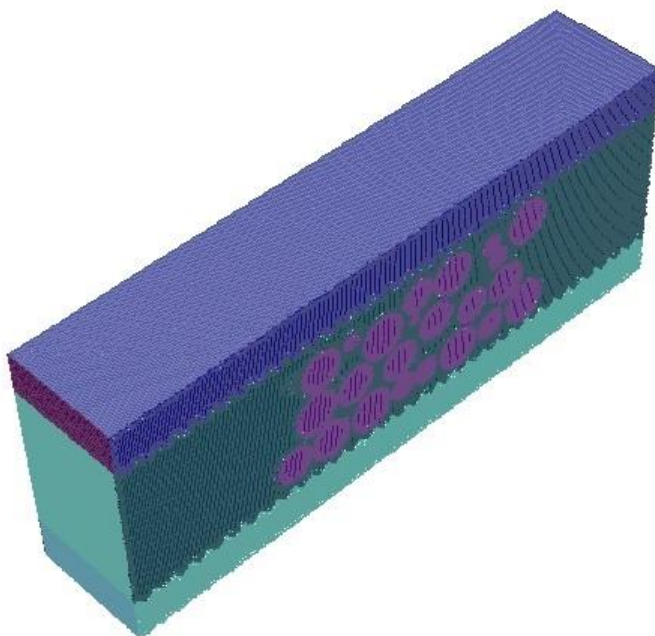


Figure 24. Meshing on geometry with corrugated fluid-fluid contact separation

Nodes	11128068	
Cells	3855009	
Element Quality	Minimum	0
	Maximum	0.7999604
Aspect Ratio	Minimum	1.356318
	Maximum	22.22024
Skewness	Minimum	0
	Maximum	0.7999604

Table 5. Mesh Metrics for packed bed w/ corrugated fluid-fluid contact surface

The simulations are carried out under laminar flow conditions with air as the working fluid. The gas inlet velocity is set at 0.005 m/s with an inlet temperature of 400 K while the superheated steam inlet velocity is set at 0.015 m/s with a temperature of 700 K. The outlet is set as pressure outlet with zero gauge pressure while the symmetry walls are given the symmetry boundary condition. The Thermal Conditions are set as Coupled with 1 mm Wall Thickness for the conducting inner walls. The heat sink rate within the particles is set at 50 kW/m³. The pressure-velocity coupling scheme is kept as SIMPLE, spatial discretization for pressure used is Second Order, while for momentum and energy it is QUICK.

Results and discussion:

A. B. & C.

Performing the three step analysis for the beds considered in this study, we obtain the following porosity profiles:

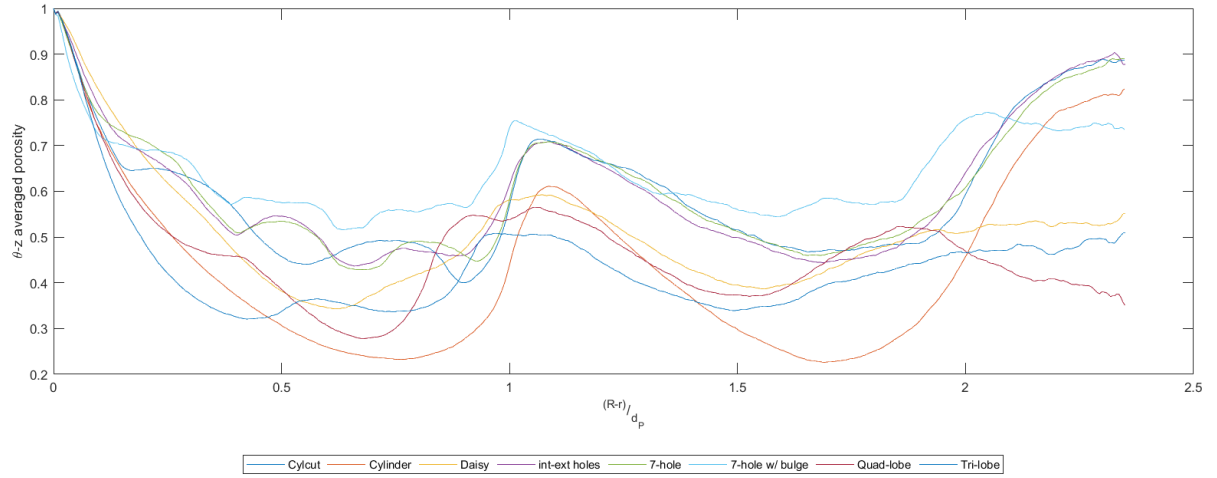


Figure 25. z-averaged radial porosity profiles for the packed beds considered in this study

We see that the z-averaged porosity values for shaped particles with internal as well as external holes is higher than particles like Cylinder and Daisy.

D. Foam bed

D1. Kelvin cell foam bed

r-z pressure contours:

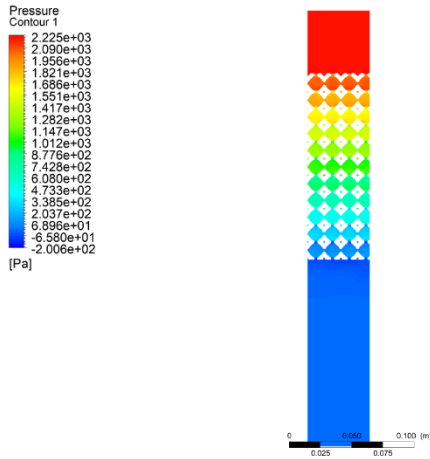


Figure 26. k- ϵ

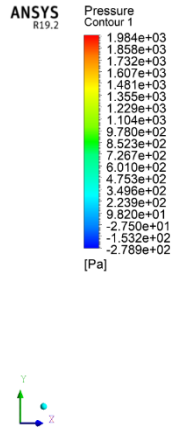


Figure 27. k- ω

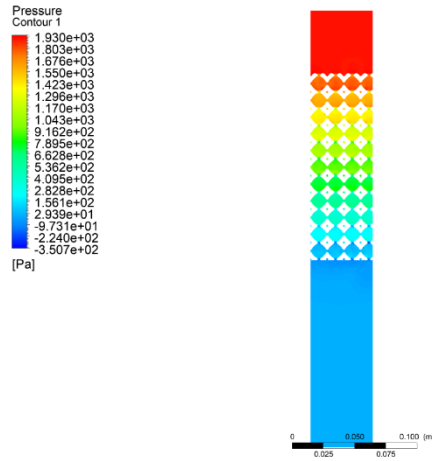
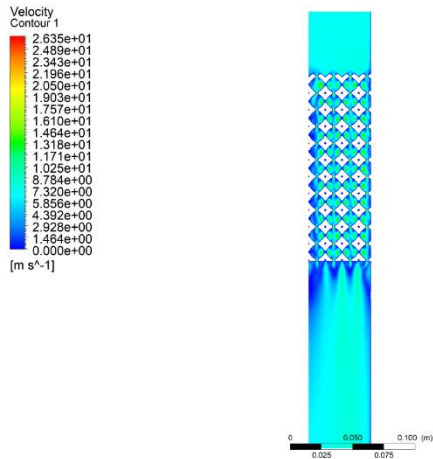
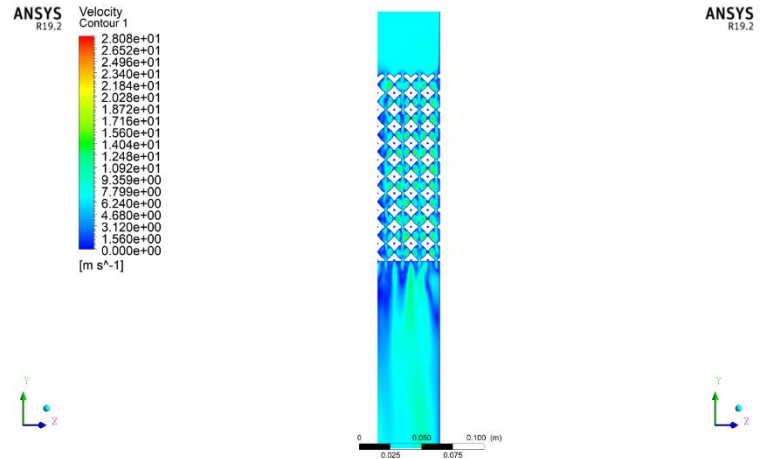
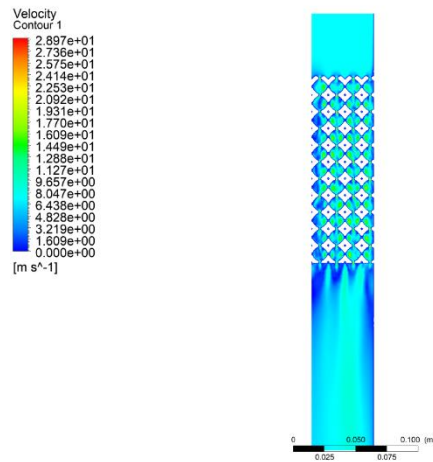
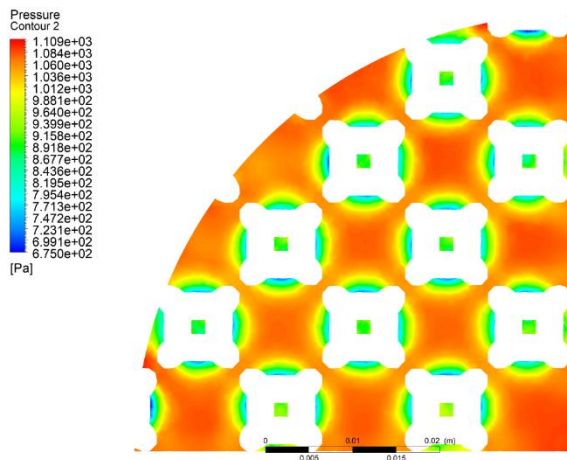
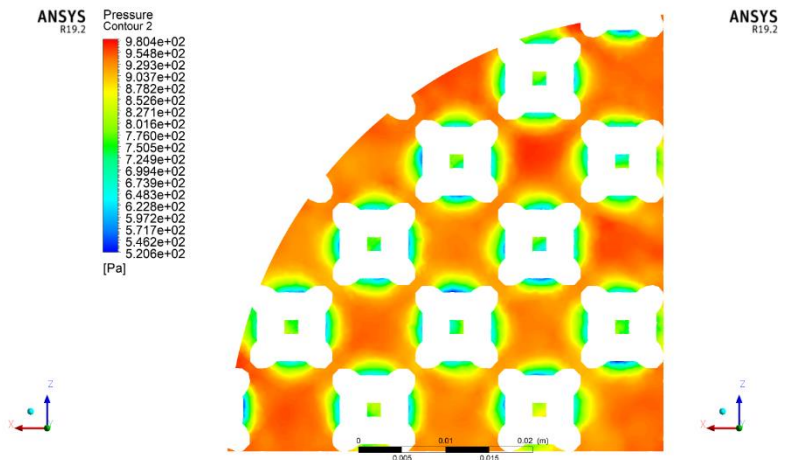


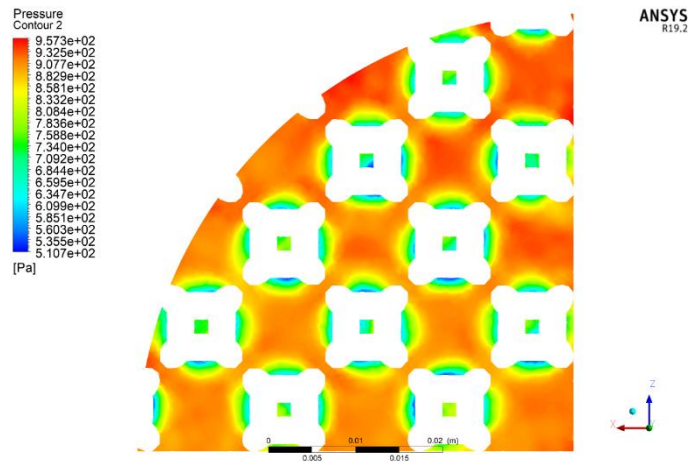
Figure 28. SST k- ω

r- θ velocity contours:

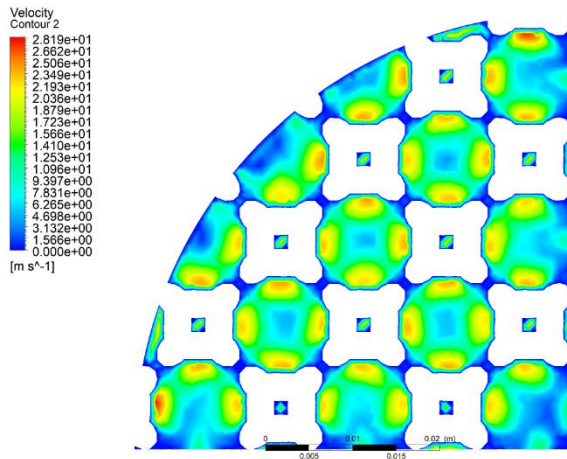
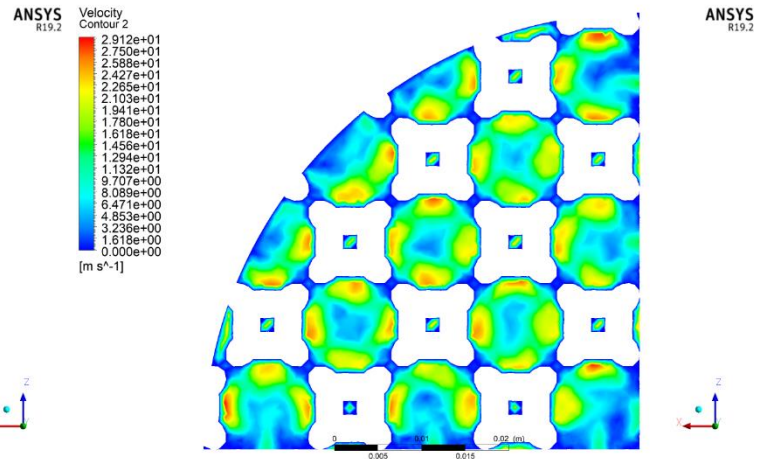
Figure 29. k- ϵ Figure 30. k- ω Figure 31. SST k- ω

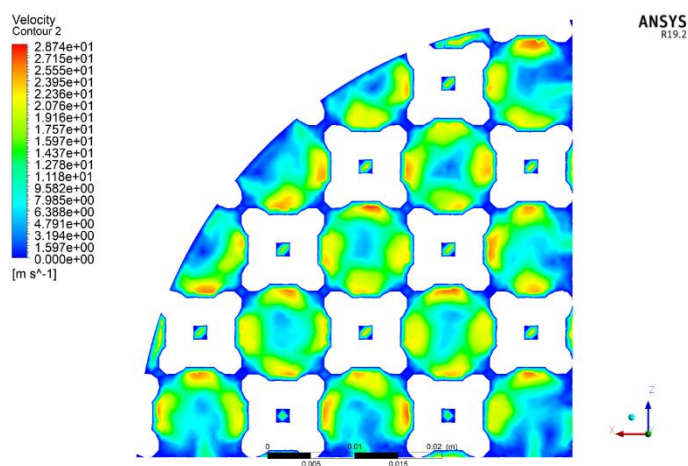
r- θ pressure contours at z = 125 mm:

Figure 32. k- ϵ Figure 33. k- ω

Figure 34. SST $k-\omega$

$r-\theta$ velocity contours at $z = 125$ mm:

Figure 35. $k-\epsilon$ Figure 36. $k-\omega$

Figure 37. SST k- ω

The pressure and velocity contours for the three different turbulence models are very similar qualitatively. As can be seen in the r- θ contours and the r-z velocity contour, the flow quantities do not vanish to zero near the symmetry walls as is expected.

D2. Spherical particle foam bed

1% volume expanded spherical particle bed

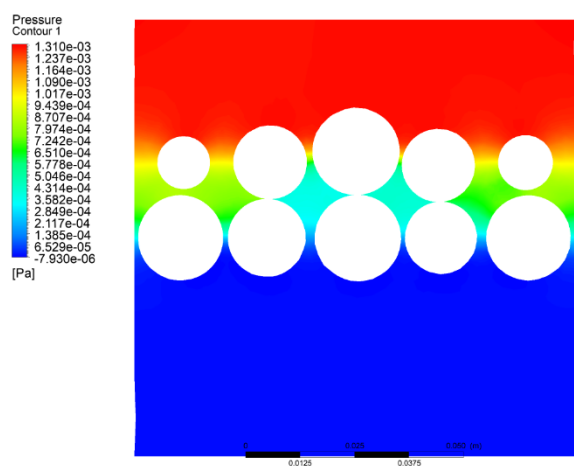


Figure 38. r-z pressure contour

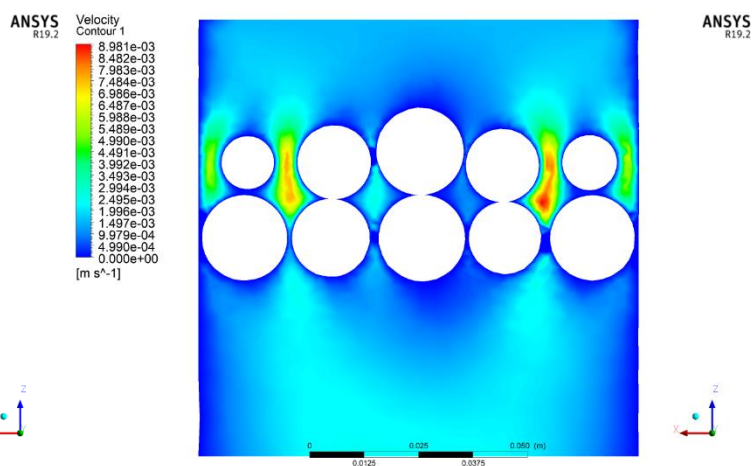


Figure 39. r-z velocity contour

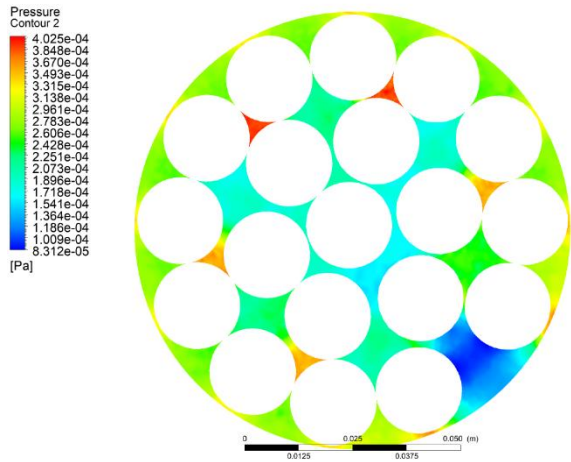


Figure 40. r- θ pressure contour
at $z = 50$ mm

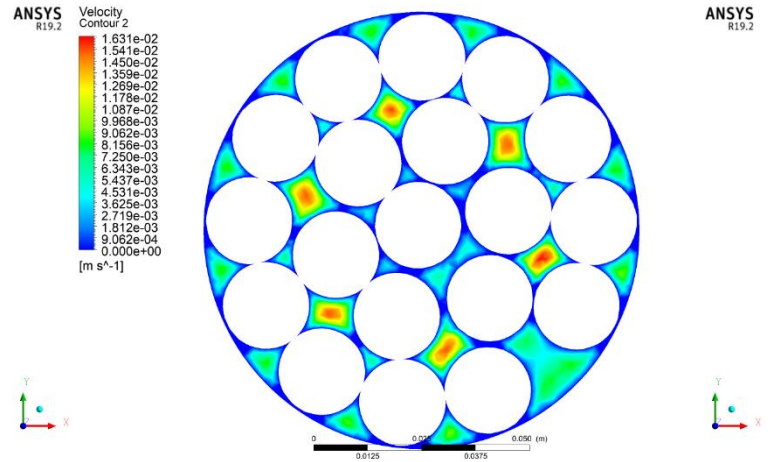


Figure 41. r- θ velocity contour
at $z = 50$ mm

5% volume expanded spherical particle bed

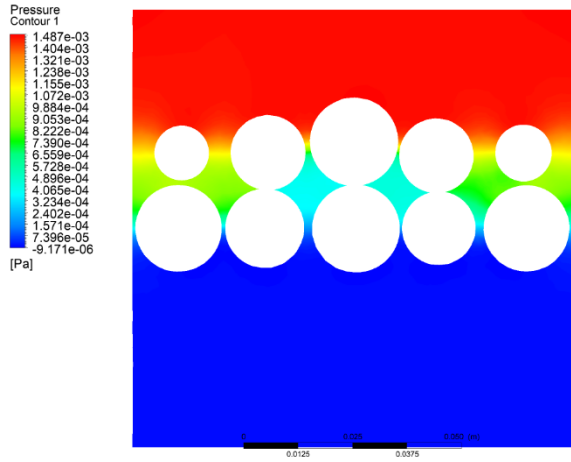


Figure 42. r- z pressure contour

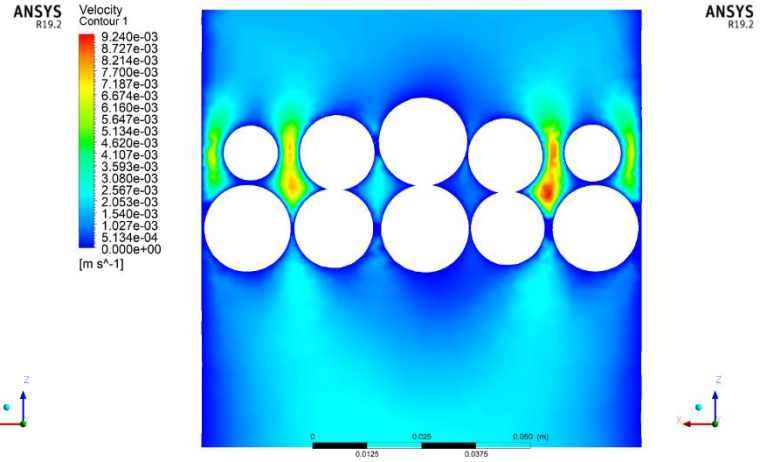


Figure 43. r- z velocity contour

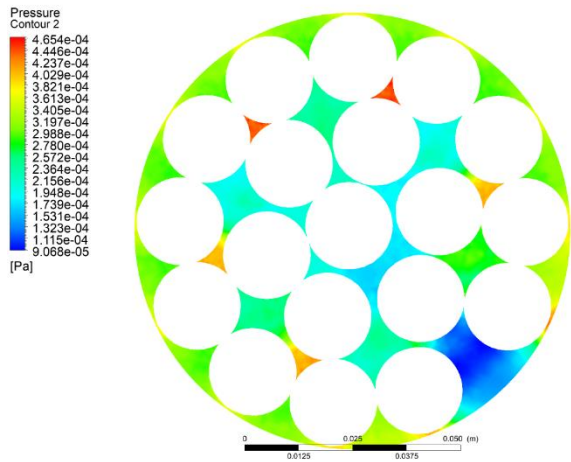


Figure 44. r- θ pressure contour
at $z = 50$ mm

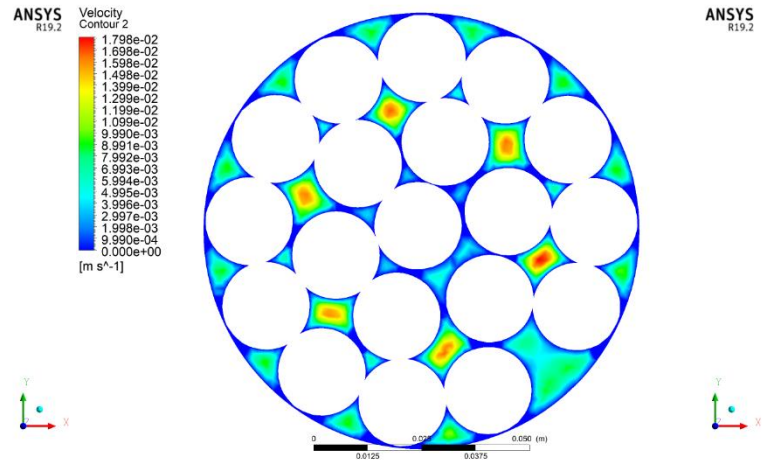


Figure 45. r- θ velocity contour
at $z = 50$ mm

10% volume expanded spherical particle bed

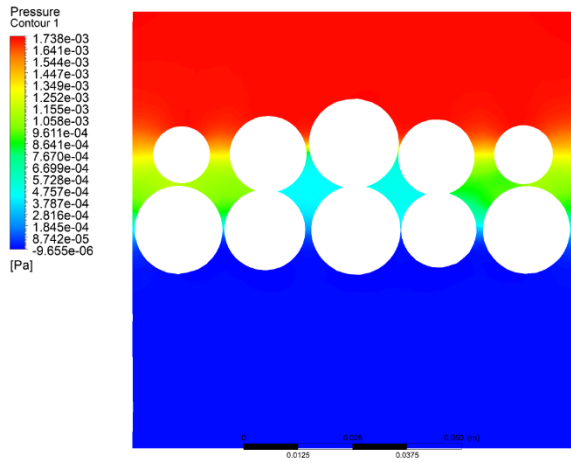


Figure 46. r-z pressure contour

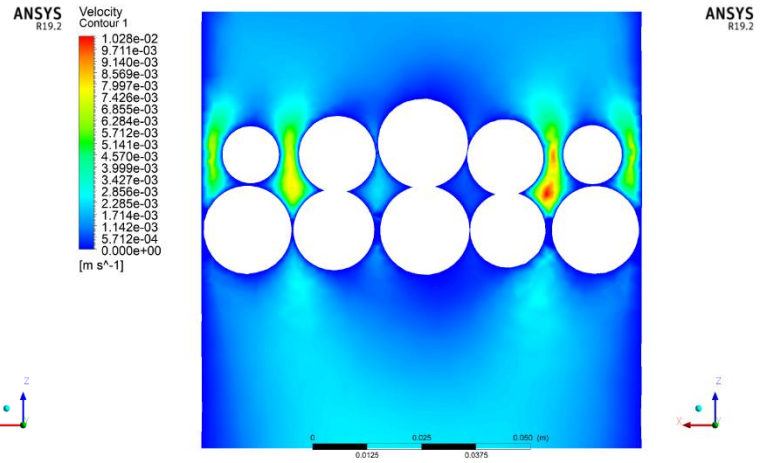


Figure 47. r-z velocity contour

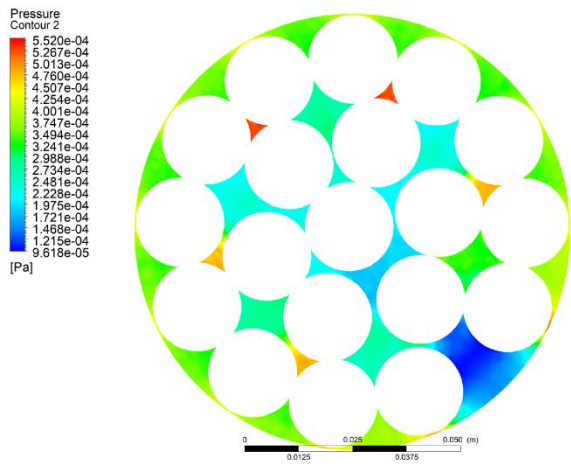


Figure 48. r- θ pressure contour
at $z = 50$ mm

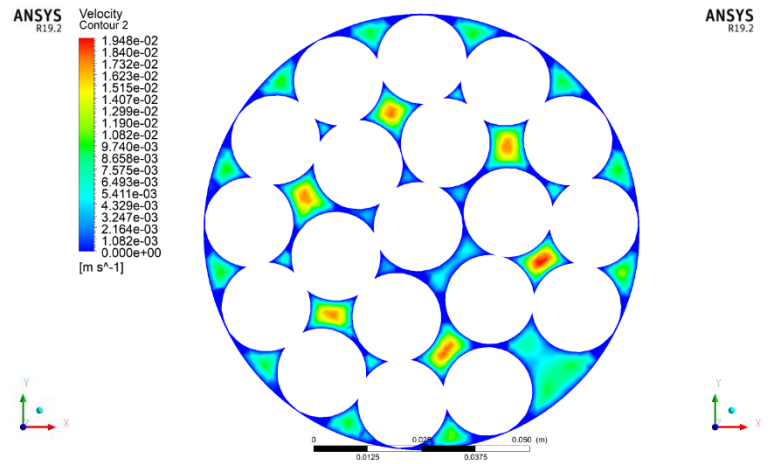


Figure 49. r- θ velocity contour
at $z = 50$ mm

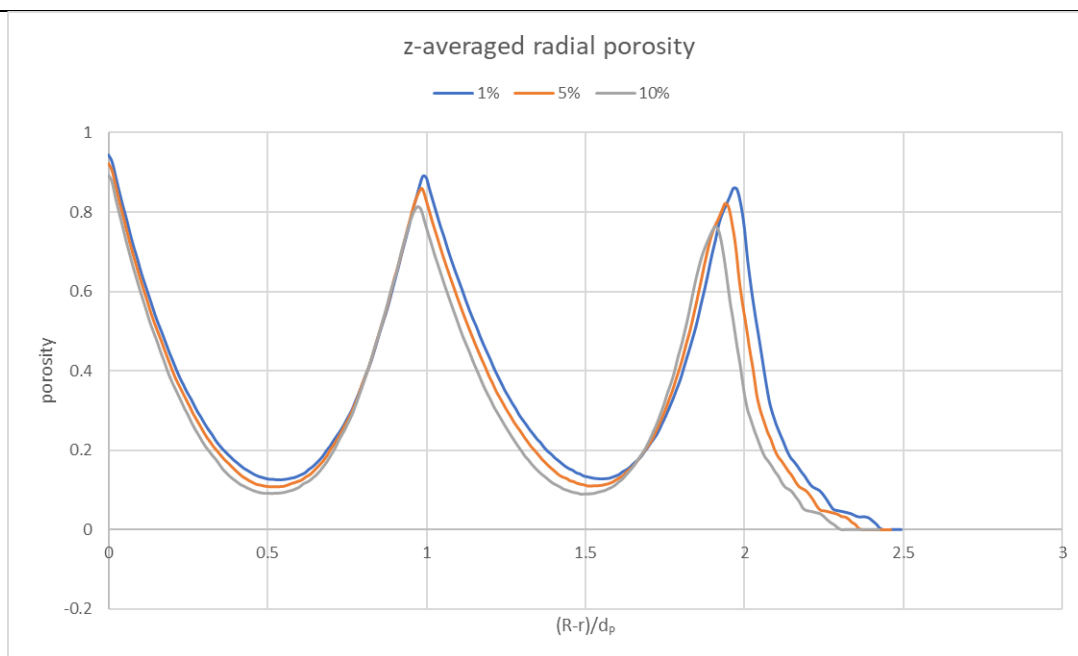


Figure 50. z-averaged radial porosity profile

The following images show heat transfer simulation results for the spherical particle foam bed with modifications made to the geometry as discussed earlier:

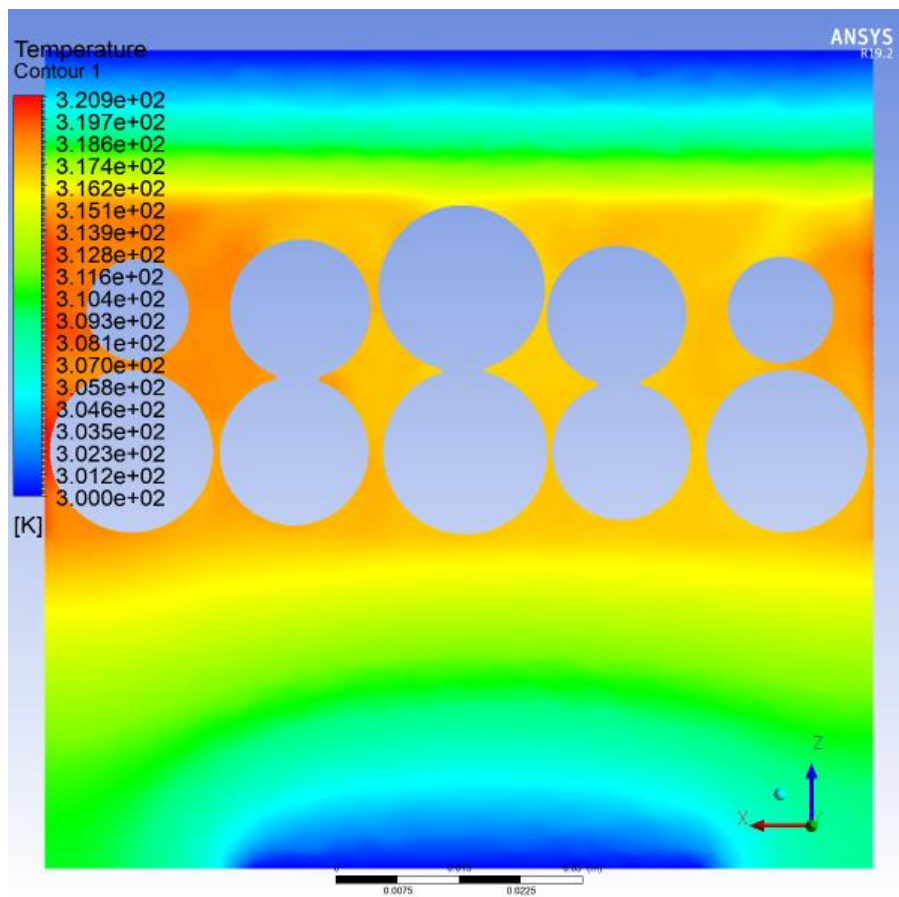


Figure 51. r-z temperature contour

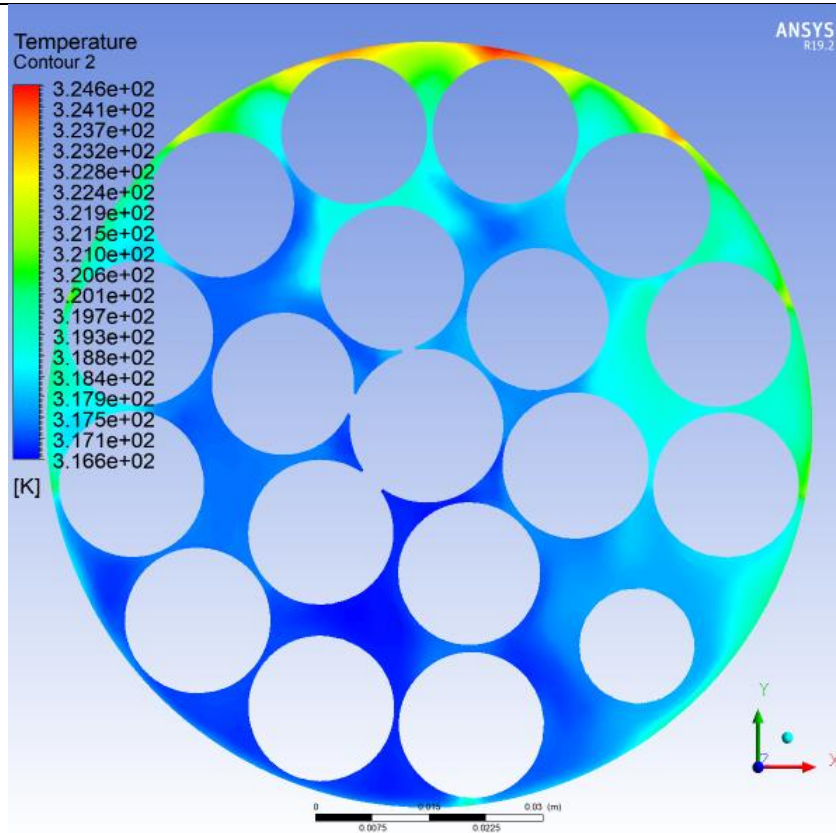


Figure 52. r- θ temperature contour at $z = 28.5$ mm

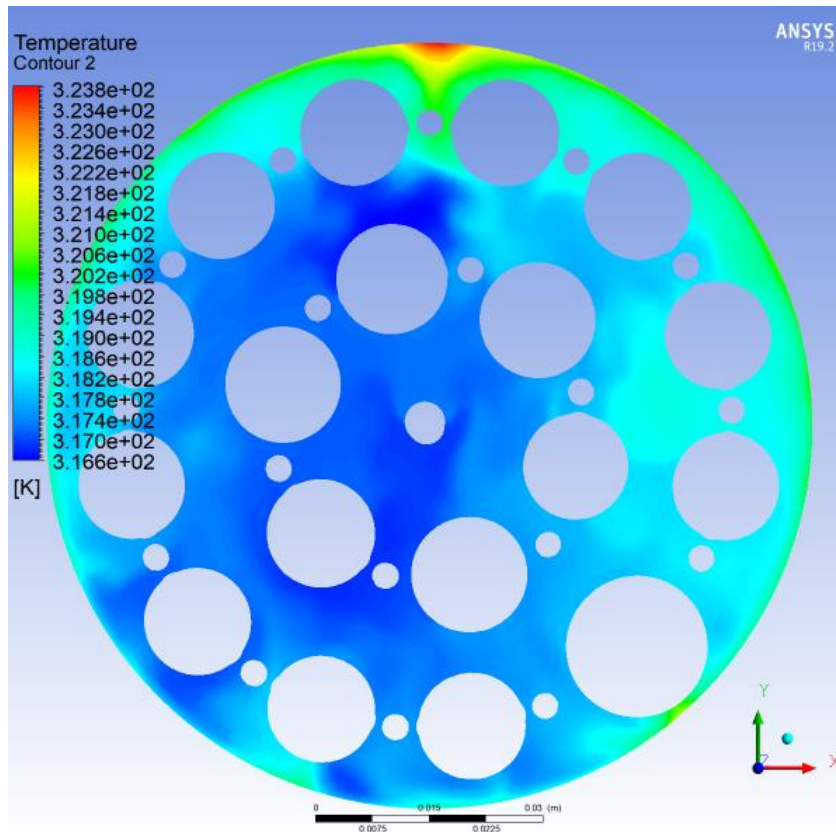


Figure 53. r- θ temperature contour at $z = 39$ mm

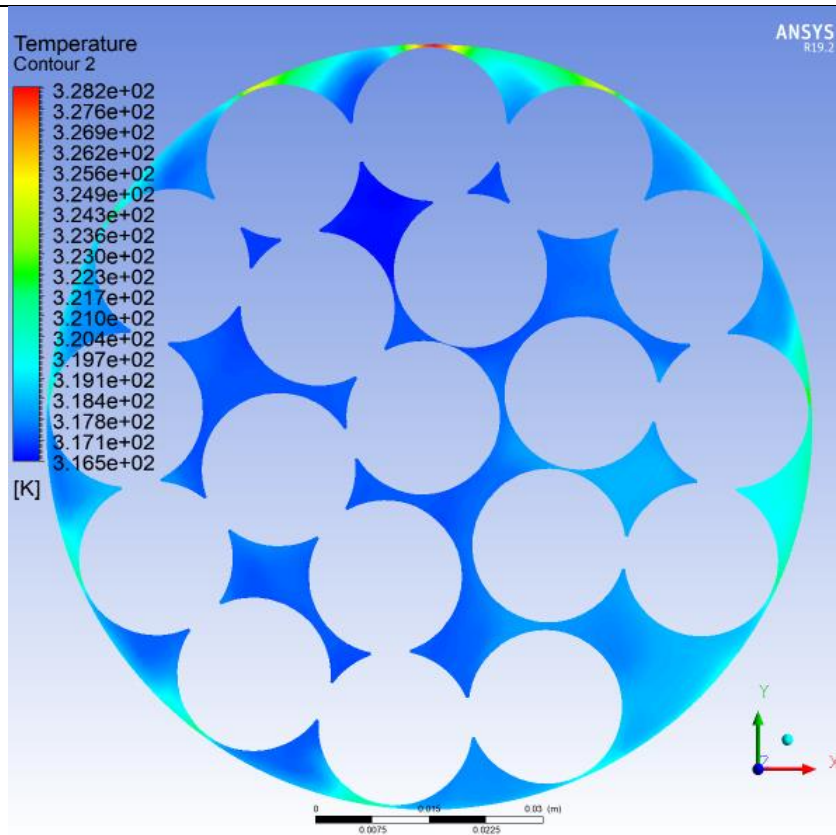


Figure 54. r- θ temperature contour at $z = 49.5$ mm

E. Packed bed

The temperature contours for the spherical particle packed with planar contact surface between the fluid-fluid contact regions are given below in Figures 55-57.

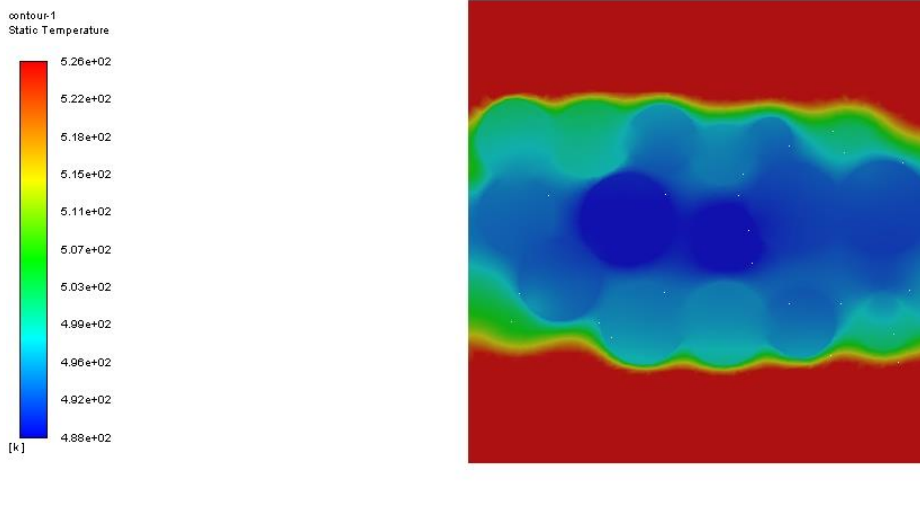


Figure 55. x-y temperature contour at $z = 47.675$ mm

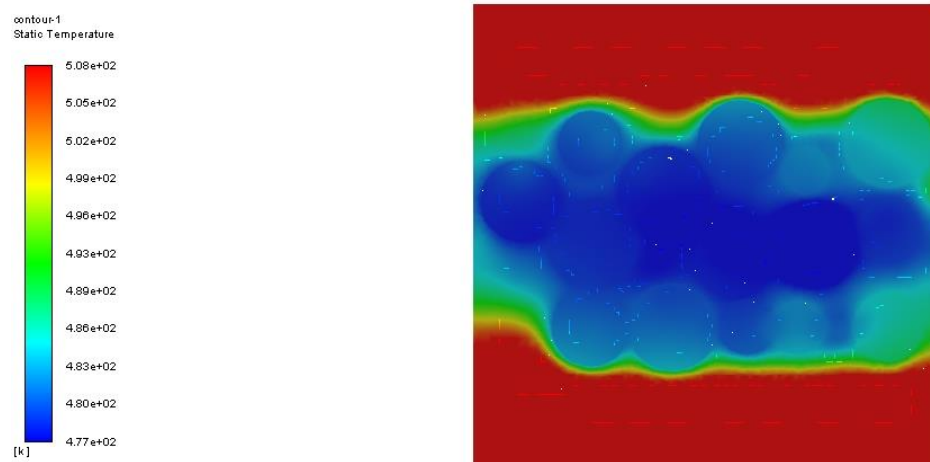


Figure 56. x-y temperature contour at $z = 65.35$ mm

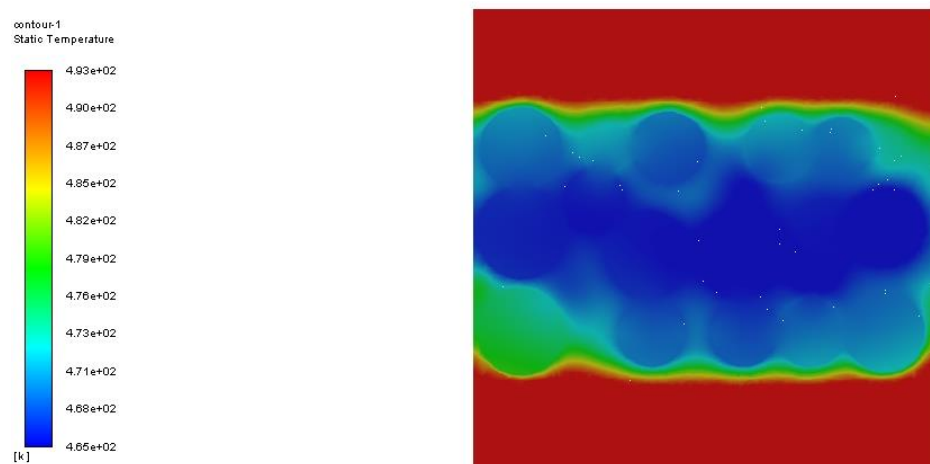


Figure 57. x-y temperature contour at $z = 83.025$ mm

Table 6 below shows the heat balance through the surfaces of the spherical particle packed bed with planar fluid-fluid contact surface.

	Heat Transfer Rate (W)
Gas inlet	0.28881374
Lower steam inlet	3.9397629
Upper steam inlet	3.9392901
Gas outlet	1.7387055
Lower steam outlet	1.9216263
Upper steam outlet	1.9120423
Upper conducting wall	2.027248
Lower conducting wall	2.0181342
Symmetry walls	0
Non-conducting walls (2)	0
Particle sink	2.5953277

Table 6. Heat balance for the spherical particle packed bed with planar fluid-fluid contact surface

Figures 58-60 display the temperature contours for the spherical particle packed bed with corrugated contact surface at the fluid-fluid contact.

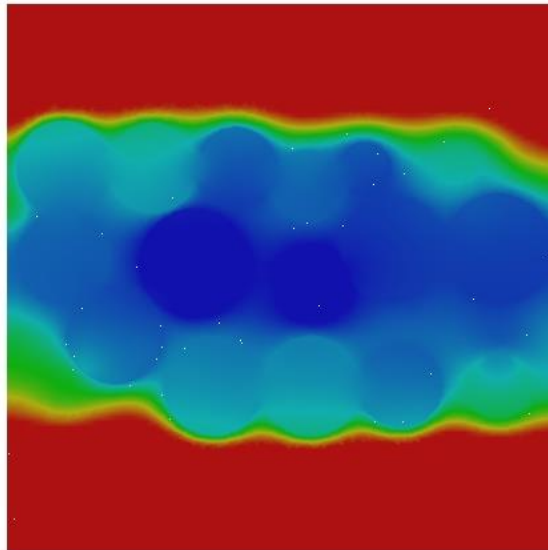
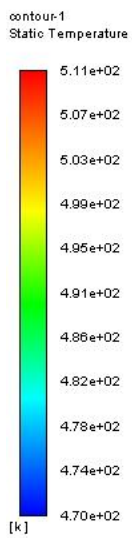


Figure 58. x-y temperature contour at $z = 47.675$ mm

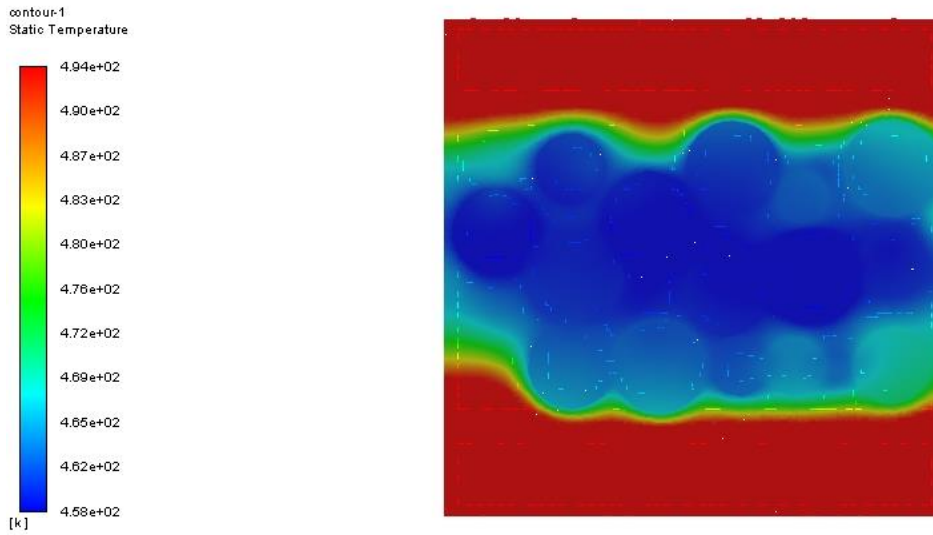


Figure 59. x-y temperature contour at $z = 65.35$ mm

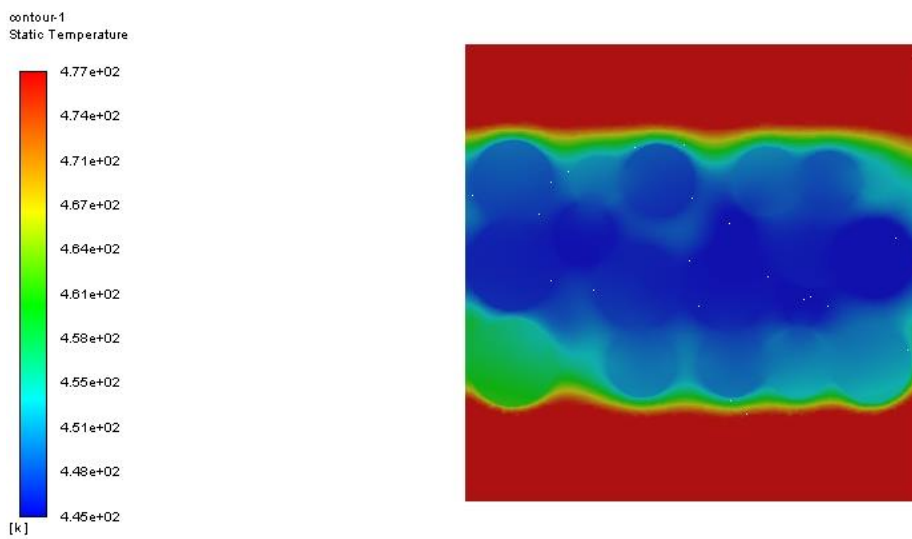


Figure 60. x-y temperature contour at $z = 83.025$ mm

Table 6 below shows the heat balance through the surfaces of the spherical particle packed bed with corrugated fluid-fluid contact surface.

	Heat Transfer Rate (W)
Gas inlet	0.08218884
Lower steam inlet	4.0424649
Upper steam inlet	3.5120993
Gas outlet	1.692286
Lower steam outlet	1.7550149
Upper steam outlet	1.4029687
Upper conducting wall	2.1091243
Lower conducting wall	2.2874399
Symmetry walls	0
Non-conducting walls (2)	0
Particle sink	2.6219564

Table 7. Heat balance for the spherical particle packed bed with corrugated fluid-fluid contact surface

References:

1. Partopour, B., Dixon, A. G. (2017). An integrated workflow for resolved-particle packed bed models with complex particle shapes. Powder Technology, 332, 258-272. <https://doi.org/10.1016/j.powtec.2017.09.009>
2. Partopour, B.; Dixon, A.G. Effect of particle shape on methanol partial oxidation in a fixed bed using CFD reactor modeling. AICHE J. 2020, 66, e16904.
3. Roshani, Shahram (1990) Elucidation of local and global structural properties of packed bed configurations. PhD thesis, University of Leeds.
4. Karthik GM, Buwa VV. Structure-resolved CFD simulations of different catalytic structures in a packed bed. Ind Eng Chem Res. 2019; 58(49): 22363- 22375.
5. Christopher Rumsey. NASA Langley Research Centre - Turbulence Modeling Resource
6. Menter, F. R., Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications. AIAA Journal, Vol. 32, No. 8, August 1994, pp. 1598-1605. <https://doi.org/10.2514/3.12149>
7. Apsley, D., Leschziner, M. Advanced Turbulence Modelling of Separated Flow in a Diffuser. Flow, Turbulence and Combustion 63, 81 (2000). <https://doi.org/10.1023/A:1009930107544>